

European Option Pricing with Lévy Processes

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1 Introduction

Modeling financial returns requires frameworks capable of capturing empirical features such as heavy tails, skewness, and abrupt price movements. Classical diffusion-based models such as Black–Scholes fail to capture empirical features like jumps, skewness and heavy tails observed in financial returns. This limitation motivates the use of Lévy processes, which provide a flexible framework incorporating discontinuities while preserving tractability.

Lévy processes provide a natural and mathematically tractable extension of diffusion models by allowing for jumps of varying magnitude and frequency, while preserving key properties such as stationary and independent increments. Within this class, models such as the Variance Gamma and Merton jump-diffusion have become standard tools for representing return dynamics and improving option pricing accuracy.

This work investigates the extent to which Lévy-based models can effectively capture the statistical and distributional properties of financial returns, and how these properties translate into option pricing implications. In particular, we focus on the interplay between model specification, parameter estimation, and numerical pricing methods.

The analysis is structured around three main components. First, we present the theoretical foundations of Lévy processes, with particular emphasis on their characteristic function representation and the Lévy–Khinchine formula. Second, we study two benchmark models, the Variance Gamma model and the Merton jump-diffusion, and examine parameter estimation techniques, including maximum likelihood estimation and the empirical characteristic function approach. Special attention is given to robustness and numerical stability in the estimation procedures. Third, we apply these models to the pricing of European options using Fourier-based methods, in particular the Carr–Madan approach, and assess their empirical performance.

Beyond a purely technical implementation, this work aims to provide a coherent framework linking probabilistic modeling, statistical estimation, and computational methods. This integrated perspective allows for a deeper understanding of the strengths and limitations of Lévy-based models when confronted with real financial data.

2 Lévy Processes

Lévy processes provide a natural extension of the Black–Scholes model by introducing trajectories that combine a Brownian component, jumps of small or large magnitude, and a deterministic drift. Unlike the classical BS model, which assumes continuous log-normal returns and constant volatility, Lévy processes capture features observed in financial markets such as sudden price jumps, heavy tails, and skewed return distributions (Baurdoux and Papapantoleon 2015; Schilling 2014; Kiesel 2014).

In the context of European call pricing, this generalisation offers several advantages: a more realistic representation of return distributions, allowing better option valuation when markets anticipate extreme events; and the possibility to calibrate the model to implied volatilities across maturities and strikes, improving the fit of the volatility surface.

2.1 Definition of Lévy processes

Lévy processes are stochastic processes characterized by stationary and independent increments, together with càdlàg paths, forming a natural extension of Brownian motion.

Based on Baurdoux and Papapantoleon 2015 (*2. Definition of Levy processes*), a stochastic process $(X_t)_{t \geq 0}$ is called a *Lévy process* if the following conditions hold:

1. **Independent increments:** for any $0 \leq t_1 < t_2 < \dots < t_n$, the increments

$$X_{t_2} - X_{t_1}, X_{t_3} - X_{t_2}, \dots, X_{t_n} - X_{t_{n-1}}$$

are mutually independent.

2. **Stationary increments:** for all $0 \leq s < t$,

$$X_t - X_s \stackrel{d}{=} X_{t-s},$$

i.e. the distribution of the increment depends only on the length of the time interval.

3. **Stochastic continuity:** for any $\varepsilon > 0$,

$$\lim_{h \rightarrow 0} \mathbb{P}(|X_{t+h} - X_t| > \varepsilon) = 0.$$

These properties imply that there exists a modification of X whose paths are **càdlàg**, i.e. right-continuous with left limits.

In particular, for such a version, the following hold no deterministic jumps for any $t \geq 0$,

$$\mathbb{P}(X_t = X_{t-}) = 1.$$

2.2 Lévy–Itô Decomposition

Any Lévy process can be decomposed into a drift, a Brownian component and a jump part, according to the Lévy–Itô decomposition Schilling 2014 (*equation 1.5*).

$$X_t = \gamma t + \sigma W_t + \int_0^t \int_{|x| < 1} x \tilde{N}(ds, dx) + \int_0^t \int_{|x| > 1} x N(ds, dx) \quad (1)$$

where $W_t \sim N(0, t)$ is a standard Brownian motion, $N(ds, dx)$ with intensity $ds \nu(dx)$ is a Poisson random measure, $\tilde{N}(ds, dx) = N(ds, dx) - ds \nu(dx)$ is the compensated measure for small jumps ($|x| < 1$), $\gamma \in \mathbb{R}$ and $\sigma \geq 0$ are respectively the drift and the volatility of the Brownian component.

ν is the *Lévy measure*, defined on $\mathbb{R} \setminus \{0\}$ and satisfying :

$$\int_{\mathbb{R}} (1 \wedge x^2) \nu(dx) < \infty. \quad (2)$$

The measure ν describes the frequency and distribution of jumps of size x , including the abundance of small jumps, the possible presence of rare but extreme jumps, asymmetry between upward and downward movements, and tail heaviness. In other words, the jump structure is governed by a Lévy measure $\nu(dx)$, which specifies the expected number of jumps per unit time with size in a given set, over the interval $[0, t]$.

2.3 Lévy–Khintchine Formula

The Lévy–Khintchine formula provides a complete characterization of Lévy processes via the triplet (γ, σ, ν) , capturing drift, diffusion and jump behavior. Kiesel 2014.

2.3.1 Characteristic Function and Lévy Exponent

The characteristic function enables to fully characterize their distribution through the Lévy exponent Kiesel 2014 (*1.2. Characteristic Functions*).

For any $u \in \mathbb{R}$,

$$\phi_{X_t}(u) = \mathbb{E}[e^{iuX_t}] = e^{t\psi(u)}. \quad (3)$$

with $\psi(u)$ the *Lévy exponent* defined by

$$\psi(u) := \log \mathbb{E}[e^{iuX_1}],$$

2.3.2 Component decomposition

By considering X_t as our equation 1 we can decomposed each element of the characteristic function 3:

$$\mathbb{E}\left[\exp(iu\gamma t)\right] = \exp(iu\gamma t) \quad (4)$$

$$\mathbb{E}\left[\exp(iu\sigma W_t)\right] = \exp\left(-\frac{1}{2}\sigma^2 u^2 t\right) \quad (5)$$

$$\mathbb{E}\left[\exp\left(iu\left(\int_0^t \int_{|x|<1} x \tilde{N}(t, dx) + \int_0^t \int_{|x|>1} x N(ds, dx)\right)\right)\right] = \exp\left(t \int_{\mathbb{R}} (e^{iux} - 1 - iux \mathbf{1}_{|x|<1}) \nu(dx)\right) \quad (6)$$

Combining all components we can obtain the Lévy exponent :

$$\psi(u) = iu\gamma - \frac{1}{2}\sigma^2 u^2 + \int_{\mathbb{R}} (e^{iux} - 1 - iux \mathbf{1}_{\{|x|<1\}}) \nu(dx) \quad (7)$$

The deterministic part is represented by equation 4. The Brownian motion as represented in equation 5 with σ the continuous fluctuation (*Derivations of the basic characteristic functions of normal distribution, are collected in Appendix A.2.* The jumps process are represented by equation 6, which will be detailed in the next part *Compound Poisson Process in Lévy*. The triplet (γ, σ, ν) fully characterises the process: drift, continuous fluctuations, and jumps.

Thus, the Lévy–Khintchine formula is defined as :

$$\phi_{X_t}(u) = \exp\left[t\left(iu\gamma - \frac{1}{2}\sigma^2 u^2 + \int_{\mathbb{R}} (e^{iux} - 1 - iux \mathbf{1}_{\{|x|<1\}}) \nu(dx)\right)\right]. \quad (8)$$

Theorem 30 in Kiesel 2014

2.4 Compound Poisson Processes in Lévy

Compound Poisson processes constitute a fundamental class of Lévy processes used to model jump components in asset prices (Kiesel 2014 (1.3.2. *Poisson Process*)).

The Poisson process is a counting process satisfying the condition of stochastic process in Definition 2.1.

$$\mathbb{P}(N_t = k) = e^{-\lambda t} \frac{(\lambda t)^k}{k!} \quad (9)$$

It models the random occurrence of events over time ($N_t = k$), where λ represents the average number of occurrence per unit time.

The Compound Poisson Processes extends this model by associating a random size $\mathbb{R}^d$ to each jump, defined as :

$$X_t = \sum_{i=1}^{N_t} Y_i \quad (10)$$

Where $(N_t)_{t \geq 0}$ Poisson process with intensity λ , and $(Y_i)_{i \geq 1}$ are i.i.d random variables with distribution G and are also independent of (N_t) .

This construction models a system where each event occurs randomly in time, and each event carries a random magnitude.

For the following step, derivations of the basic characteristic functions of the Compound Poisson Process, are collected in Appendix A.1 if needed.

2.4.1 Large Jumps

Let us define $\phi_{J_t^{(>1)}}(u)$, the characteristic function for large-jump component of a Lévy process $J_t^{(>1)}$ such as :

$$J_t^{(>1)} = \int_0^t \int_{|x|>1} x N(ds, dx) \quad (11)$$

We consider the set of a large jumps, defined by $|x| > 1$. Since the Lévy measure satisfies $\nu(x \in \mathbb{R} : |x| > 1) < \infty$, the number of such jumps over any finite time interval is finite.

$$N([0, t] \times \{x \in \mathbb{R} : |x| > 1\})$$

$$N_1^{(>1)} \sim \text{Poisson}(\nu(|x| > 1)).$$

Conditionally on the number of jumps, the corresponding jump amplitudes $(Y_i)_{i \geq 1}$ are independent and identically distributed with law given by the normalized restriction of the Lévy measure :

$$\mathbb{P}(Y_i \in dx) = \frac{\nu(dx)}{\nu(|x| > 1)} \quad (12)$$

As a consequence, the large-jump component of the Lévy process can be written as a compound Poisson process :

$$X_t^{(>1)} = \sum_{i=1}^{N_t^{(>1)}} Y_i \quad (13)$$

Thus, the final expression of the characteristic function $\phi_{J_t^{(>1)}}(u)$ is:

$$\phi_{J_t^{(>1)}}(u) = \exp \left(t \int_{|x|>1} (e^{iux} - 1) \nu(dx) \right) \quad (14)$$

Theorem 34 in Kiesel 2014

2.4.2 Small Jumps and Compensation

Let us define $\phi_{J_t^{(<1)}}(u)$, the characteristic function for small-jump component of a Lévy process $J_t^{(<1)}$ such as :

$$J_t^{(<1)} = \int_0^t \int_{|x|<1} x \tilde{N}(ds, dx), \quad (15)$$

If $\nu(|x| < 1) < \infty$, then only finitely many small jumps occur on any finite interval, and this term can be treated as a compound Poisson process.

However, if $\nu(|x| < 1) = \infty$, there are almost surely infinitely many small jumps, and the raw sum in does not converge in general. In particular, the integral may diverge due to the accumulation of infinitely many terms with non-zero expectation.

To resolve this issue, one introduces a compensated Poisson integral by subtracting the expected contribution of small jumps:

$$J_t^{(<1)} = \int_{|x|<1} x \tilde{N}(ds, dx) = \int_{|x|<1} x (N(ds, dx) - t\nu(dx)), \quad (16)$$

where $\tilde{N}(ds, dx) = N(ds, dx) - t\nu(dx)$. This removes the divergent linear drift induced by the infinite activity of small jumps.

The corresponding characteristic function of the (non-compensated) small-jump component is:

$$\mathbb{E}\left[e^{iu \int_{|x|<1} x N(ds, dx)}\right] = \exp\left(t \int_{|x|<1} (e^{iux} - 1) \nu(dx)\right). \quad (17)$$

To analyze the behavior of this integral, we use the Taylor expansion of the exponential function around $x = 0$:

$$e^{iux} - 1 = iux - \frac{1}{2}u^2x^2 - \frac{i}{6}u^3x^3 + \dots$$

For small values of $|x|$, higher-order terms become negligible, and the dominant term is iux . Therefore,

$$e^{iux} - 1 \approx iux \quad \text{as } x \rightarrow 0.$$

If $\nu(|x| < 1) = \infty$, the integral behaves like:

$$\int_{|x|<1} iux \nu(dx),$$

which may diverge. This shows that the linear term is responsible for the divergence.

To ensure convergence, one subtracts this linear term and considers:

$$e^{iux} - 1 \longrightarrow e^{iux} - 1 - iux \mathbf{1}_{|x|<1}.$$

Then, for small x ,

$$e^{iux} - 1 - iux = -\frac{1}{2}u^2x^2 + O(x^3),$$

so the integrand behaves like x^2 near zero.

Since the Lévy measure satisfies 2, the characteristic function becomes :

$$\phi_{J_t^{(<1)}}(u) = \exp\left(t \int_{|x|<1} (e^{iux} - 1 - iux) \nu(dx)\right) \quad (18)$$

Hence, the integral is finite, which guarantees convergence of the characteristic exponent and ensures that the compensated small-jump component is well-defined with càdlàg paths.

Theorem 28 in Kiesel 2014

3 Specific Models (Variance Gamma, Merton)

Several parametric Lévy models have been proposed to better reproduce the statistical properties of financial returns.

3.1 Variance Gamma

The Variance Gamma process is a pure-jump Lévy process obtained by subordinating Brownian motion with a Gamma process (see Madan and Seneta 1990; Carr, Madan, and Chang 1998). The study of the VG process provides a concrete example of a Lévy process with an explicitly computable characteristic function, whose parameters are directly linked to the Lévy triplet.

The Variance Gamma (VG) process can be defined via *subordination* as a Brownian motion evaluated at a random time change given by a Gamma process. Following the original construction in Madan and Seneta 1990 (see Eq. (7)), the VG process is written in structural form as

$$X_t = b(G_t), \quad (19)$$

where $\{G_t\}_{t \geq 0}$ is a Gamma process and $b(\cdot)$ denotes a Brownian motion with drift θ and volatility σ .

Equivalently, using the canonical decomposition of Brownian motion,

$$b(s) = \theta s + \sigma W_s, \quad (20)$$

the VG process admits the following analytical representation:

$$X_t = \theta G_t + \sigma W_{G_t}. \quad (21)$$

where, $(W_s)_{s \geq 0}$ is a standard Brownian motion, $(G_t)_{t \geq 0}$ is a Gamma subordinator $G_t \sim \Gamma(\alpha = t/\nu, \beta = \nu)$, independent of W . , $\theta \in \mathbb{R}$ is a drift parameter, $\sigma, \nu > 0$ are respectively the volatility associated with the Brownian component and the jump-activity parameter.

Both formulations are strictly equivalent. While the structural form is conceptually fundamental, the analytical form is generally preferred in practical derivations and applications.

3.1.1 Pricing approaches

The VG process is a Lévy process of finite variation and infinite activity, with no Brownian component. Given estimated parameters (θ, σ, ν) , there are two equivalent approaches to pricing European options.

- **Via the Lévy measure $\nu(dx)$:**

The characteristic function can be constructed using the Lévy–Khintchine formula

$$\varphi_{X_T}(u) = \exp \left[iu\gamma T + T \int_{\mathbb{R}} (e^{iux} - 1 - iux\mathbf{1}_{|x| < 1}) \nu(dx) \right] \quad (22)$$

where the Lévy measure $\nu(ds)$ of the VG process is given explicitly (see Carr, Madan, and Chang 1998, Eq. (14)) by

$$\nu(dx) = \frac{\exp\left(\frac{\theta x}{\sigma^2}\right)}{|x|} \exp\left(-\frac{\sqrt{2\sigma^2/\nu + \theta^2}}{\sigma^2}|x|\right) dx. \quad (23)$$

This expression follows from the infinite activity jump structure inherited from the gamma subordinator.

- **Directly via the VG characteristic function:**

Alternatively, the characteristic function admits the closed-form expression (see Carr, Madan, and Chang 1998 Eq. (7)):

$$\varphi_{X_T}(u) = \left(1 - i\theta\nu u + \frac{1}{2}\sigma^2\nu u^2 \right)^{-T/\nu}. \quad (24)$$

Both approaches are mathematically equivalent. The first emphasizes the Lévy structure of the process and relies on the explicit form of the Lévy measure. However, it requires evaluating a non-trivial integral in the Lévy–Khintchine representation, it require to integrate a modified Bessel function (see Carr, Madan, and Chang 1998 Eq. (23)). In contrast, the second expression is obtained by conditioning on the gamma time change and exploiting the Gaussian structure of the subordinated Brownian motion. This leads directly to a closed-form characteristic function. For this reason, the analytical expression is significantly more tractable in practice, and is typically preferred for option pricing via Fourier methods such as the Carr–Madan transform.

3.1.2 Characteristic Function

Let us defined the characteristic function of X_t as $\rho_{X_t}(u) = \mathbb{E}[e^{iuX_t}]$.

Conditionally on $G_t = g$:

$$\mathbb{E}[e^{iu(\theta g + \sigma W_g)}] = e^{iu\theta g - \frac{1}{2}\sigma^2 u^2 g} = e^{gA}, \quad A = iu\theta - \frac{1}{2}\sigma^2 u^2.$$

Since $W_g \sim \mathcal{N}(0, g)$, this follows from the standard property of Brownian motion.

Expectation with respect to G_t . Because $G_t \sim \Gamma(t/\nu, \nu)$,

$$\mathbb{E}[e^{AG_t}] = (1 - \nu A)^{-t/\nu}, \quad \Re(A) < 1/\nu.$$

Thus the characteristic function of the VG process is:

$$\rho_{X_t}(u) = \left(1 - \nu \left(iu\theta - \frac{1}{2}\sigma^2 u^2\right)\right)^{-t/\nu} \quad (25)$$

see Appendix A.2 and A.3 for demonstration of the characteristic function for normal and gamma distribution.

3.1.3 Interpretation of the Parameters

ν controls jump frequency, with large ν signifiante high jump activity. σ adjusts jump magnitude, a larger σ produces larger jumps. And θ controls skewness, the sign and direction of asymmetry.

Since subordination implies $\mathbb{E}[G_t] = t$, we have $\alpha\beta = t$ with $\alpha = t/\nu$, $\beta = \nu$.

As $\nu \rightarrow 0$,

$$\lim_{\nu \rightarrow 0} (1 - \nu A)^{-t/\nu} = e^{tA},$$

and the VG process converges to a Brownian motion with drift θ and volatility σ .

3.2 The Merton Model

The Merton jump-diffusion model extends the Black–Scholes framework by incorporating log-normal jumps, allowing for skewness and excess kurtosis in asset returns Merton 1976.

3.2.1 Definition of the Merton Process

The log-price $X_t = \ln S_t$ satisfies Merton 1976 Eq. (2):

$$X_t = \left(\mu - \frac{\sigma^2}{2} - \lambda k\right)t + \sigma W_t + \sum_{i=1}^{N_t} Y_i, \quad (26)$$

where : $k = \mathbb{E}[e^{Y_i} - 1] = e^{\mu_J + \frac{1}{2}\sigma_J^2} - 1$.

W_t is a standard Brownian motion, σ is the diffusive volatility, $N_t \sim \text{Poisson}(\lambda t)$ is the number of jumps, $Y_i \sim \mathcal{N}(\mu_J, \sigma_J^2)$ is the i -th jump size, γ compensates for drift induced by jumps so that the expected return remains μ .

According to Matsuda 2004, the drift $\gamma = \alpha - \frac{\sigma^2}{2} - \lambda k$ is derived to ensure the process is a martingale under the risk-neutral measure. This is further elaborated in the discussion surrounding Equation (6) and the Lévy-Khintchine representation in Equation (8), which provides the characteristic function and cumulant generating function for the model.

3.2.2 Characteristic Function

Let us defined the characteristic function of X_t as $\phi_{X_t}(u) = \mathbb{E}[e^{iuX_t}]$

The process decomposes into a continuous part and a jump part:

$$X_t = X_t^{\text{cont}} + X_t^{\text{jumps}}.$$

Brownian part. The continuous component is normal :

$$\mathbb{E}[e^{iuX_t^{\text{cont}}}] = \exp\left(iu\left(\mu - \frac{\sigma^2}{2} - \lambda k\right)t - \frac{1}{2}\sigma^2 u^2 t\right). \quad (27)$$

Jump part. Conditionally on $N_t = n$:

$$\sum_{i=1}^n Y_i \sim \mathcal{N}(n\mu_J, n\sigma_J^2),$$

and thus

$$\mathbb{E}[e^{iuX_t^{\text{jumps}}}] = \exp\left(\lambda t(e^{iu\mu_J - \frac{1}{2}\sigma_J^2 u^2} - 1)\right). \quad (28)$$

see Appendix A.1 and A.2 for demonstration of the characteristic function for normal distribution and Poisson process.

Full characteristic function. Combining the two components (see Matsuda 2004 Eq. (8):

$$\phi_{X_t}(u) = \exp\left\{iu\left(\mu - \frac{\sigma^2}{2} - \lambda k\right)t - \frac{1}{2}\sigma^2 u^2 t + \lambda t(e^{iu\mu_J - \frac{1}{2}\sigma_J^2 u^2} - 1)\right\}. \quad (29)$$

These models illustrate different approaches to incorporating jumps in financial modeling, balancing tractability and realism.

4 Parameter Estimation and Optimization Algorithm

To estimate the parameters of the models under study, two main methods are used: maximum likelihood estimation (MLE) and a method based on the characteristic function.

4.1 Maximum Likelihood Estimation

Parameter estimation via maximum likelihood relies on maximizing the log-likelihood function associated with the observed data, a standard approach in statistical inference.

Let X be a real-valued random variable whose distribution depends on a parameter vector $\Theta \in \mathbb{R}^d$. Denote by $f(x; \Theta)$ its probability density function.

Given an i.i.d. sample $(x_1, \dots, x_n)$, the likelihood function is defined as:

$$L(x_1, \dots, x_n; \Theta) = \prod_{i=1}^n f(x_i; \Theta),$$

and the log-likelihood is:

$$\ell(\Theta) = \sum_{i=1}^n \log f(x_i; \Theta).$$

The Maximum Likelihood Estimator (MLE) is defined as:

$$\hat{\Theta} = \arg \max_{\Theta} \ell(\Theta),$$

which is computed numerically by minimizing the negative log-likelihood.

In the following, we assume $t = 1$, corresponding to one observation period (e.g. daily log-returns).

4.1.1 Variance Gamma Model

Due to the presence of special functions such as the modified Bessel function, maximum likelihood estimation for the Variance Gamma model may suffer from numerical instability).

The Variance Gamma process admits the representation:

$$X = \gamma + \theta G_t + \sigma W_{G_t}, \quad (30)$$

where $Z \sim \mathcal{N}(0, g)$ and $G \sim \Gamma(1/\nu, \nu)$.

Here, γ captures the baseline drift, while θG_t and σW_{G_t} , capture asymmetry and volatility. The marginal density is given by:

$$f_{VG}(x; \Theta) = f_{VG}(x; \gamma, \theta, \sigma, \nu),$$

with the closed-form expression involving the modified Bessel function K_ν :

$$f_{VG}(x; \theta, \sigma, \nu) = \frac{2e^{\theta x/\sigma^2}}{\sigma \sqrt{2\pi\nu^\alpha} \Gamma(\alpha)} \left(\frac{|x|}{\sqrt{\theta^2 + 2\sigma^2/\nu}} \right)^{\alpha-1/2} K_{\alpha-1/2} \left(\frac{|x|}{\sigma^2} \sqrt{\theta^2 + 2\sigma^2/\nu} \right), \quad (31)$$

where $\alpha = 1/\nu$.

The VG process is a subordinated process, where a Brownian motion with drift is evaluated at a random time given by a gamma process. This subordination leads to a density that involves the Bessel function, which arises naturally from the integral representation of the gamma process and the normal density (see the derivation in the Appendix of Carr, Madan, and Chang 1998 and Eq. (23)).

In practice, the likelihood is evaluated numerically. A direct implementation from Laptev 2016 is:

```

1 def vg_neg_loglik(params, data):
2     c, sigma, theta, nu = params
3     if sigma <= 0 or nu <= 0:
4         return 1e12
5     f = pdf(data, c=c, sigma=sigma, theta=theta, nu=nu)
6     f = np.maximum(f, 1e-300) # numerical stability
7     return -np.sum(np.log(f))

```

Due to the presence of the Bessel function, this likelihood can be numerically unstable. For this reason, alternative methods such as ECF estimation are often preferred for the VG model.

4.1.2 Merton Jump-Diffusion Model

The log-price dynamics are given by:

$$X = \gamma + \sigma W + \sum_{i=1}^N Y_i, \quad (32)$$

where $N \sim \text{Poisson}(\lambda)$ and $Y_i \sim \mathcal{N}(\mu_J, \sigma_J^2)$.

The marginal density is a Poisson mixture of Gaussian densities:

$$f_M(x; \Theta) = \sum_{n=0}^{\infty} \frac{e^{-\lambda} \lambda^n}{n!} \mathcal{N}(x; \gamma + n\mu_J, \sigma^2 + n\sigma_J^2).$$

In practice, the infinite sum is truncated at a finite level $n_{\max}$.

```

1 def merton_pdf(x, c, sigma, lam, muJ, sigmaJ, n_max=50):
2     x = np.asarray(x)
3     pdf = np.zeros_like(x, dtype=float)
4     for n in range(n_max + 1):
5         w_n = np.exp(-lam) * lam**n / factorial(n)

```

```

6         mu_n = c + n * muJ
7         var_n = sigma**2 + n * sigmaJ**2
8         pdf += w_n * norm.pdf(x, mu_n, np.sqrt(var_n))
9     return pdf

```

The negative log-likelihood is:

```

1 def merton_neg_loglik(params, data):
2     c, sigma, lam, muJ, sigmaJ = params
3     if sigma <= 0 or sigmaJ <= 0 or lam < 0:
4         return 1e12
5     f = merton_pdf(data, c, sigma, lam, muJ, sigmaJ)
6     f = np.maximum(f, 1e-300)
7     return -np.sum(np.log(f))

```

4.2 Empirical Characteristic Function Method (ECF)

An alternative approach consists in matching the empirical and theoretical characteristic functions (see Carrasco and Florens 2000 Section [2.1]), which is particularly effective when the density is not available (see Carrasco and Florens 2000 Section [1]) in closed form. This approach is particularly useful when the marginal density is complex or difficult to manipulate analytically, while the characteristic function admits a simple closed-form expression.

4.2.1 Theoretical Characteristic Function

For an increment X_t , the characteristic function is defined as

$$\phi(u; \Theta) = \mathbb{E}[e^{iuX_t}]. \quad (33)$$

Closed-form expressions are available for both the Variance Gamma model (25) and the Merton jump-diffusion model (29). These expressions are implemented in a unified manner as follows:

```

1 def phi_model(u, params, model):
2     if model == "VG":
3         c, sigma, theta, nu = params
4         base = 1 - 1j * theta * nu * u + 0.5 * sigma**2 * nu * u**2
5         return np.exp(1j * c * u) * base**(-1.0/nu)
6
7     elif model == "Merton":
8         c, sigma, lam, muJ, sigmaJ = params
9         jump = np.exp(1j*u*muJ - 0.5*sigmaJ**2 * u**2)
10        return np.exp(1j*u*c - 0.5*sigma**2*u**2 + lam*(jump - 1))

```

4.2.2 Empirical Characteristic Function

Given a sample $(r_1, \dots, r_n)$, the empirical characteristic function is defined by (see Carrasco and Florens 2000) section [2.1] :

$$\hat{\phi}(u) = \frac{1}{n} \sum_{j=1}^n e^{iur_j}. \quad (34)$$

Its numerical implementation is given by:

```

1 def ecf_empirical(u, r):
2     return np.mean(np.exp(1j * np.outer(u, r)), axis=1)

```

4.2.3 Objective Function

Parameter estimation is performed by minimizing a weighted distance between the empirical and theoretical characteristic functions, in line with generalized method of moments approaches based on a continuum of moment conditions (see Carrasco and Florens 2000, Section 4.3):

$$J(\Theta) = \sum_{k=1}^K w(u_k) \left| \hat{\phi}(u_k) - \phi(u_k; \Theta) \right|^2. \quad (35)$$

The frequency grid $(u_k)_{k=1}^K$ is constructed over a bounded interval, where the upper bound depends on a tail index in order to adapt to the heaviness of the distribution:

```

1 def make_uniform_u_grid(tail_index, N, u_min_cap=1, u_max_cap=20):
2     u_max0 = 10 / tail_index
3     u_max = np.clip(u_max0, u_min_cap, u_max_cap)
4     u = np.linspace(0, u_max, N)
5     return u, u_max

```

Low frequencies ($u \approx 0$) contain most of the information about the distribution, while high frequencies are significantly noisier. To mitigate this effect, a decreasing weighting function is introduced. Such weighting schemes are standard in characteristic-function-based estimation (see Carrasco and Florens 2000, Section 4.3):

$$w(u) = \frac{1}{1 + u^2}. \quad (36)$$

```

1 wfun = lambda u: 1.0 / (1.0 + u**2)

```

Thus, the objective function is implemented in a model-independent framework, with parameter constraints enforced to ensure admissibility:

```

1 def ecf_objective(params, u, r, model,
2                   wfun=lambda u: 1.0/(1.0 + u**2)):
3
4     if model == "VG":
5         c, sigma, theta, nu = params
6         if sigma <= 0 or nu <= 0:
7             return 1e10
8
9     elif model == "Merton":
10        c, sigma, lam, muJ, sigmaJ = params
11        if sigma <= 0 or lam < 0 or sigmaJ <= 0:
12            return 1e10
13
14        phi_hat = ecf_empirical(u, r)
15        phi_theo = phi_model(u, params, model)
16
17        diff = np.abs(phi_hat - phi_theo)**2
18        return np.sum(diff * wfun(u))

```

4.2.4 Parameter Initialization

A suitable initialization is crucial for optimization convergence. It is typically based on empirical moments or on a diffusion/jump separation.

Variance Gamma (VG). Instead of using raw moments, initial values can be defined using robust estimates:

1. **Location (c)**

Use the empirical mean of log-returns:

$$c_0 = \bar{r} = \frac{1}{n} \sum_{i=1}^n r_i.$$

2. **Diffusion volatility (σ)**

Use a robust estimate based on the MAD (Median Absolute Deviation):

$$\sigma_0 = \text{MAD} = \text{median}(|r_i - \bar{r}|),$$

adjusted by the 75% quantile of the standard normal (reduces sensitivity to extreme returns):

$$\sigma_0 = \frac{\text{MAD}}{\Phi^{-1}(0.75)}.$$

3. **Asymmetry / drift of the Gamma clock (θ)**

Initial guess based on empirical skewness, bounded to avoid extreme values:

$$\theta_0 = \text{sign}(\text{skew}) \cdot \min(|\text{skew}| \cdot \sigma_0, \sigma_0).$$

4. **Shape parameter (ν)**

Instead of $\sigma_0^2/(\text{kurtosis} - 3)$, which may blow up when kurtosis is close to 3, use a bounded version:

$$\nu_0 = \text{clip}\left(\frac{\sigma_0^2}{\max(\text{kurtosis} - 3, 0.1)}, 0.01, 10\right).$$

in python the parameter initialisation are defined as :

```

1 def automatic_init_vg(logr):
2
3     logr = np.asarray(logr)
4
5     c0 = np.mean(logr)
6
7     MAD = np.median(np.abs(logr - c0))
8     sigma0 = MAD / 0.6744897501960817 # Phi^-1(0.75)
9
10    skew = np.mean((logr - c0)**3) / sigma0**3
11    theta0 = np.sign(skew) * min(abs(skew)*sigma0, sigma0)
12
13    r_kurt = kurtosis(logr, fisher=False) # normal = 3
14    nu0 = np.clip(sigma0**2 / max(r_kurt - 3, 0.1), 0.01, 10.0)
15
16    return np.array([c0, sigma0, theta0, nu0], dtype=float)
17

```

Merton Model (normal jumps). Parameter initialization is done in several steps to separate the diffusion and jump components and remain robust to extreme values.

1. **Location:** Use the median of log-returns:

$$\hat{\mu} = \text{median}(r_1, \dots, r_n).$$

2. **Robust volatility:** Based on MAD:

$$\text{MAD} = \text{median}(|r_i - \hat{\mu}|), \quad \sigma_{\text{rob}} = \frac{\text{MAD}}{\Phi^{-1}(0.75)}.$$

3. **Jump detection:** Returns exceeding a threshold $\text{thr} \cdot \sigma_{\text{rob}}$ are considered jumps:

$$\mathcal{J} = \{r_i : |r_i - \hat{\mu}| > \text{thr} \cdot \sigma_{\text{rob}}\}, \quad \text{thr} = 3.$$

4. **Jump frequency and parameters:** Empirical jump frequency and jump moments:

$$\hat{\lambda} = \frac{|\mathcal{J}|}{n}, \quad \hat{\lambda} \geq 10^{-6},$$

$$(\hat{\mu}_J, \hat{\sigma}_J) = \begin{cases} \left(\frac{1}{|\mathcal{J}|} \sum_{r_i \in \mathcal{J}} r_i, \sqrt{\frac{1}{|\mathcal{J}| - 1} \sum_{r_i \in \mathcal{J}} (r_i - \hat{\mu}_J)^2} \right), & |\mathcal{J}| \geq 2, \\ (-0.02, 0.05), & \text{otherwise (fallback)}. \end{cases}$$

5. **Diffusion volatility:** Estimated on the full sample:

$$\hat{\sigma} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (r_i - \bar{r})^2}, \quad \text{if } \hat{\lambda} > 0.05 : \quad \hat{\sigma} \leftarrow \max(0.8\hat{\sigma}, 10^{-6}).$$

```

1 def automatic_init_merton(logr, thr=3.0):
2
3     logr = np.asarray(logr)
4     c0 = np.median(logr)
5     sigma_rob = 1.4826 * np.median(np.abs(logr - c0))
6     jumps = logr[np.abs(logr - c0) > thr * sigma_rob]
7     lam0 = max(len(jumps)/len(logr), 1e-6)
8
9     if len(jumps) >= 2:
10         muJ0 = np.mean(jumps)
11         sigmaJ0 = np.std(jumps, ddof=1)
12         sigmaJ0 = max(sigmaJ0, 1e-3)
13     else:
14         muJ0 = -0.02 if c0 >= 0 else 0.02
15         sigmaJ0 = max(0.05, 0.5*sigma_rob)
16     sigma0 = np.std(logr, ddof=1)
17     if lam0 > 0.05:
18         sigma0 = max(0.8*sigma0, 1e-6)
19
20     return np.array([c0, sigma0, lam0, muJ0, sigmaJ0], dtype=float)

```

These estimation methods provide complementary approaches depending on the tractability of the model.

4.3 Optimization Problem and Nelder–Mead Method

The optimization problem is solved using the Nelder–Mead simplex algorithm, a derivative-free method well suited for nonlinear and potentially non-smooth objective functions Nelder and Mead 1965.

The ECF estimator is defined as:

$$\hat{\Theta} = \arg \min_{\Theta} J(\Theta),$$

and is obtained numerically using a derivative-free optimization method.

4.3.1 Problem Formulation

Whether using MLE or ECF, parameter estimation amounts to solving

$$f(\Theta) = \begin{cases} -\log L(\Theta) & (\text{MLE}), \\ \sum_{k=1}^m w_k |\hat{\phi}(u_k) - \phi(u_k; \Theta)|^2 & (\text{ECF}). \end{cases}$$

The function f is generally nonlinear, multimodal, and may be nondifferentiable, justifying the use of a gradient-free method such as the Nelder–Mead algorithm.

4.3.2 Principle of the Nelder–Mead Method

The ECF estimate is:

$$\hat{\Theta} = \arg \min_{\Theta} J(\Theta).$$

Nelder–Mead (Wikipedia (2025)) uses a *simplex* of $n + 1$ points in a space of dimension n .

- **Simplex:** set of $n+1$ points (e.g. 4 points for VG).
- **Operations:** reflection, expansion, contraction (inner or outer), reduction.

The idea is to move the simplex to explore the parameter space:

- *Reflection* tests the direction opposite the worst point;
- *Expansion* extends a promising direction;
- *Contraction* brings the simplex closer to the best point when no improvement is found;
- *Reduction* shrinks the simplex around the best solution.

Simplex Initialization

1. Choose an initial point x_0 (from moments or a robust estimate).
2. Construct n near points to form a simplex.
3. Evaluate $f(x_i)$ and sort the points:

$$f(x_1) \leq f(x_2) \leq \dots \leq f(x_{n+1}).$$

Simplex Operators

- **Centroid:**

$$x_c = \frac{1}{n} \sum_{i=1}^n x_i.$$

- **Reflection:**

$$x_r = x_c + \alpha(x_c - x_h), \quad \alpha > 0.$$

- **Expansion:**

$$x_e = x_c + \gamma(x_r - x_c), \quad \gamma > 1.$$

- **Contraction (outer or inner):**

$$x_{co} = x_c + \beta(x_r - x_c), \quad x_{ci} = x_c - \beta(x_c - x_h), \quad 0 < \beta < 1.$$

- **Reduction:**

$$x_i = x_1 + \delta(x_i - x_1), \quad 0 < \delta < 1.$$

4.3.3 Criteria and Statistical Tests

To evaluate the fit of the Variance Gamma (VG) and Merton models, we use the following criteria:

- **Kolmogorov–Smirnov (KS):** maximum distance between the empirical distribution F_n and the theoretical distribution F :

$$D_n = \sup_x |F_n(x) - F(x; \hat{\Theta})|.$$

- **Wasserstein Distance (order 1):** mean transport distance between two distributions P and Q :

$$W_p(\mu, \nu) = \inf \left\{ \left(\mathbb{E}[d(X, Y)^p] \right)^{1/p} \mid \mathbb{P}_X = \mu, \mathbb{P}_Y = \nu \right\}.$$

5 Pricing of a European Call

Option pricing in Lévy models is typically performed under a risk-neutral measure to ensure the absence of arbitrage Carr and Madan 1999; Madan and Seneta 1990.

5.1 Price of a European Call

We consider a financial asset $(S_t)_{t \geq 0}$ defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, whose underlying dynamics are modeled by a Lévy process. We work in a general framework where the risk-free rate r and the cost of carry (or dividend yield) q are constant (for this part see Carr and Madan 1999 section [3] and Madan and Seneta 1990 section [IV]).

Definition of the European Call. A European call option with maturity T and strike price K is a derivative product whose payoff at maturity is:

$$\text{Payoff} = (S_T - K)^+ = \max(S_T - K, 0).$$

The buyer of the call therefore holds the right, but not the obligation, to purchase the underlying asset at price K at time T .

Pricing Principle: Absence of Arbitrage. In the absence of arbitrage opportunities, the present value of the call must equal the expected value of its payoff, discounted at the risk-free rate. To express this principle rigorously, we introduce the fundamental notion of a *risk-neutral measure*.

Risk-Neutral Measure and Martingale Property

In an arbitrage-free market, option prices can be expressed as discounted expectations under a risk-neutral measure, ensuring that the discounted asset price is a martingale.

Under these assumptions, there exists a probability measure $\mathbb{Q}$ equivalent to the historical measure $\mathbb{P}$ such that the discounted price of the underlying becomes a martingale:

$$e^{-(r-q)t} S_t \text{ is a martingale under } \mathbb{Q}.$$

Equivalently,

$$\mathbb{E}^{\mathbb{Q}}[S_T \mid \mathcal{F}_t] = e^{(r-q)(T-t)} S_t. \quad (37)$$

This condition determines the drift under the risk-neutral measure. If the underlying follows a Lévy process,

$$S_t = S_0 e^{X_t},$$

the process (X_t) must be adjusted under $\mathbb{Q}$ so as to satisfy the martingale condition above. For the particular case of the Variance Gamma model, this leads to an explicit relation between historical and risk-neutral parameters.

General Pricing Formula for the European Call

Under the risk-neutral measure $\mathbb{Q}$, and imposing the martingale condition, the price of the European call is:

$$C(S_0, K, T) = e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+].$$

Expanding the payoff:

$$C(S_0, K, T) = e^{-rT} \int_0^\infty (s - K)^+ f_{S_T}^{\mathbb{Q}}(s) ds, \quad (38)$$

where $f_{S_T}^{\mathbb{Q}}$ denotes the density of the underlying asset price at maturity under $\mathbb{Q}$.

This formula states that the call price equals the expected value of its future gains. It's computed in a fictitious world where all assets grow at the risk-free rate, and then discounted at rate r .

Under the historical measure $\mathbb{P}$, the underlying carries a risk premium. But under the risk-neutral measure $\mathbb{Q}$, all risk premia disappear, and only the risk-free rate remains. This risk neutralization enables the expectation-based computations used in Fourier methods, in particular in the Carr–Madan method. When closed-form pricing formulas are not available, transform methods provide efficient numerical alternatives.

5.2 Carr–Madan Method

When the density of the underlying asset is not available in closed form, Fourier transform methods provide an efficient alternative by exploiting the characteristic function (for this part, see Carr and Madan 1999 section 3.1 for transform derivation and section 4 for FFT implementation.)

5.2.1 Motivation for Fourier Methods

In Lévy models such as the Variance Gamma (VG) model, the density $f_{S_T}^{\mathbb{Q}}$ is generally not available in closed form. This prevents direct integration of the payoff, which requires explicit knowledge of $f_{S_T}^{\mathbb{Q}}$. However, the characteristic function of the log-price,

$$X_T = \ln(S_T), \quad \varphi_{X_T}(u) = \mathbb{E}^{\mathbb{Q}}[e^{iuX_T}],$$

is always available in closed form for Lévy models.

Let $k = \ln(K)$ denote the log-strike. The call price can be written such as equation 38

- The payoff grows asymptotically like e^x as $x \rightarrow +\infty$. A Fourier transform requires sufficient integrability (controlled growth).
- The Carr–Madan approach introduces a damping factor ($\alpha > 0$) to ensure integrability of the payoff, enabling fast computation of option prices via the Fast Fourier Transform Carr and Madan 1999.

$$C_\alpha(k) = e^{\alpha k} C(k).$$

This damping imposes the condition

$$\mathbb{E}^{\mathbb{Q}}[S_T^{\alpha+1}] < \infty, \quad \text{equivalently} \quad \mathbb{E}^{\mathbb{Q}}[e^{(\alpha+1)X_T}] < \infty.$$

- The Carr–Madan Fourier transform is then given by

$$\psi_\alpha(\nu) = \int_{-\infty}^{+\infty} e^{i\nu k} C_\alpha(k) dk = \frac{e^{-rT} \varphi_{X_T}(\nu - i(\alpha + 1))}{\alpha^2 + \alpha - \nu^2 + i\nu(2\alpha + 1)}.$$

The proof is detailed in Appendix A.4.

- The damped price is then recovered through inverse Fourier transform:

$$C_\alpha(k) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{-i\nu k} \psi_\alpha(\nu) d\nu.$$

5.3 Integrability Conditions and the Risk–Neutral Measure for the Variance–Gamma Model

In order to apply Fourier-based pricing methods consistently within the Variance Gamma framework, it is necessary to ensure both the existence of exponential moments and the validity of the risk–neutral measure, which jointly impose constraints on the model parameters (see Carr and Madan 1999 section [5] and Carr, Madan, and Chang 1998 section [3]).

5.3.1 Existence of Exponential Moments

Under the Variance–Gamma model, the log–return process X_t has characteristic function:

$$\varphi_{X_t}(u) = \left(1 - i\theta\nu u + \frac{1}{2}\sigma^2\nu u^2\right)^{-t/\nu}.$$

The moment generating function (MGF) is obtained by substituting $u \mapsto -iu$:

$$\mathbb{E}[e^{uX_t}] = \left(1 - \theta\nu u + \frac{1}{2}\sigma^2\nu u^2\right)^{-t/\nu}.$$

It is well defined only if

$$\boxed{1 - \theta\nu u + \frac{1}{2}\sigma^2\nu u^2 > 0.}$$

This condition for the existence of exponential moments plays a central role in the Carr–Madan method as well as in the construction of the risk–neutral measure.

5.3.2 Condition on the Damping Parameter α (Carr–Madan)

The Carr–Madan method requires the existence of the exponential moment

$$\mathbb{E}\left[e^{(\alpha+1)X_T}\right] < \infty,$$

which imposes that the MGF be valid at $u = \alpha + 1$, i.e.

$$\boxed{1 - \theta\nu(\alpha + 1) + \frac{1}{2}\sigma^2\nu(\alpha + 1)^2 > 0.}$$

This condition ensures convergence of the Fourier transform of the regularized payoff and numerical stability of the FFT.

5.3.3 Risk–Neutral Measure and Computation of the Correction Term

To price options, one must work under a risk–neutral measure $\mathbb{Q}$ such that the discounted underlying price is a martingale (see Carr, Madan, and Chang 1998 section [3], Eq. (22)):

$$e^{-(r-q)t}S_t \text{ is a martingale under } \mathbb{Q}.$$

Writing $S_t = S_0 e^{X_t}$, this condition is equivalent to

$$\mathbb{E}^{\mathbb{Q}}[e^{X_t}] = e^{(r-q)t}.$$

A standard approach is to introduce a deterministic correction term w_t and write

$$\ln S_t = \ln S_0 + (r - q)t + X_t - w_t.$$

This yields

$$e^{-(r-q)t}S_t = S_0 e^{X_t - w_t}, \quad \text{and the martingale condition becomes } \mathbb{E}^{\mathbb{Q}}[e^{X_t - w_t}] = 1.$$

For a stationary Lévy process, one has the representation

$$\mathbb{E}[e^{uX_t}] = e^{tk(u)},$$

where $k(u)$ is the cumulant exponent. For the VG model,

$$k(u) = -\frac{1}{\nu} \ln(1 - \theta\nu u + \frac{1}{2}\sigma^2\nu u^2).$$

The condition $\mathbb{E}[e^{X_t - wt}] = 1$ gives

$$e^{t(k(1)-w)} = 1 \implies w = k(1) = -\frac{1}{\nu} \ln(1 - \theta\nu + \frac{1}{2}\sigma^2\nu).$$

Thus, the log-underlying dynamics under the risk-neutral measure are:

$$X_t^{\mathbb{Q}} = X_t - wt.$$

Condition for the Existence of the Unit Moment

The correction term w is well defined only if

$$1 - \theta\nu + \frac{1}{2}\sigma^2\nu > 0.$$

This is exactly the condition for the existence of the exponential moment $\mathbb{E}[e^{X_t}]$, consistent with the condition imposed on the damping parameter α in the Carr–Madan method. This consistency ensures the global validity of the mathematical framework (existence of the MGF, convergence of the FFT, absence of arbitrage).

6 Numerical Application: Pricing an SPX European Call

The numerical implementation relies on Fourier-based pricing combined with calibrated Lévy models, a standard approach in computational finance.

6.1 Construction of the Frequency Grid for the ECF and the Carr–Madan Method

Estimating the parameters of the Variance–Gamma and Merton models via the ECF, as well as pricing European options using the Carr–Madan method, requires defining a frequency grid

$$u = \{u_0, u_1, \dots, u_{N-1}\},$$

that is sufficiently rich to capture tail behaviour while remaining compatible with the constraints of the Fast Fourier Transform (FFT).

Estimation of the Tail Exponent ξ via the Hill Estimator

Let $(X_t)_{t=1}^n$ denote a series of log-returns. We consider the absolute values $|X_t|$ in order to capture both tails symmetrically. After sorting in decreasing order,

$$X_{(1)} \geq X_{(2)} \geq \dots \geq X_{(n)},$$

we select k extreme observations. The Hill estimator of the tail parameter ξ is given by:

$$\hat{\xi} = \frac{1}{k} \sum_{i=1}^k \ln\left(\frac{X_{(i)}}{X_{(k+1)}}\right).$$

A small $\hat{\xi}$ signals heavier tails, implying slower decay of the characteristic function and the need to extend the frequency grid.

Definition of the Maximum Frequency Bound

The estimated tail exponent is used to set an upper bound for the frequency grid by the heuristic:

$$u_{\max} = \text{clip}\left(\frac{10}{\hat{\xi}}, 1, 20\right),$$

to maintain a balance between:

- sufficiently covering the tails (high frequencies required when $\hat{\xi}$ is small),
- excluding very high-frequency regions where the empirical CF becomes strongly noisy.

Frequency Grid for the ECF

The final grid is defined as

$$u_k = k \Delta u, \quad k = 0, 1, \dots, N-1, \quad u_k \leq u_{\max},$$

where Δu is imposed by the FFT via the classical relation $\Delta u \Delta x = \frac{2\pi}{N}$. Only the frequencies satisfying $u_k \leq u_{\max}$ are used in the ECF estimation.

6.2 Parameters and Calibration

6.2.1 Estimation Results (Variance Gamma vs Merton)

Parameters	MLE	ECF
c	0.000916	0.000807
σ	0.011431	0.012259
θ	-0.000570	-0.000471
ν	1.296766	4.189263

Table 1: Estimated VG model parameters

Statistic	Data	MLE	ECF
KS statistic	-	0.029628	0.129648
Wasserstein	-	0.000644	0.001903
Mean	0.000337	0.000454	0.000200
Std. deviation	0.012297	0.011541	0.012298
Skewness	-0.476663	-0.076712	-0.728582
Kurtosis	12.871057	3.397900	13.485884

Table 2: Statistical comparison for the VG model

Parameters	MLE	ECF
c	0.001215	0.000587
σ	0.005919	0.008389
λ	0.368159	0.067692
μ_J	-0.002387	-0.003699
σ_J	0.016712	0.034358

Table 3: Estimated Merton model parameters

Statistic	Data	MLE	ECF
KS statistic	-	0.030026	0.061444
Wasserstein	-	0.000581	0.000936
Mean	0.000337	0.000388	0.000342
Std. deviation	0.012297	0.011579	0.011827
Skewness	-0.476663	-0.371951	-0.664946
Kurtosis	12.871057	3.945544	10.316793

Table 4: Statistical comparison for the Merton model

Variance–Gamma (VG) Model MLE and ECF produce notably different parameter estimates, especially for the shape parameter ν . The mean and standard deviation are relatively similar across methods, but the ECF slightly overestimates volatility. KS and Wasserstein distances show that MLE fits the empirical distribution of log–returns much more accurately.

MLE thus reproduces the general shape of the distribution better, especially the moderate tails, providing a more coherent fit to the overall data. The ECF tends to misrepresent the central and intermediate distribution regions despite achieving a kurtosis closer to the empirical value.

Graphical comparisons are given in Appendix B.1.

Merton Model Differences between MLE and ECF are also present for the Merton model. MLE estimates a higher jump frequency λ with more moderate jump parameters, better capturing jump behaviour in the data. KS and Wasserstein distances confirm this observation. Simulated moments from MLE reproduce empirical skewness and kurtosis more faithfully, whereas ECF underestimates jump frequency and overestimates jump dispersion.

Graphical comparisons are given in Appendix B.2.

General Comparison and Conclusion Comparing the two models and two estimation methods, we retain the Variance–Gamma parameters obtained via Maximum Likelihood Estimation (MLE) as the reference for pricing our European option.

6.2.2 Price via FFT Carr–Madan

For the numerical example, to compute a 6-month European call option, we consider:

- Spot: $S_0 = 6870.399$,
- Maturity: $T = 108/365 \approx 0.296$ years,
- Risk-free rate: $r = 3.83\%$, *U.S. Department of the Treasury (2025)*
- Dividend yield: $q = 0$,
- VG parameters: $\theta = -0.000564$, $\sigma = 0.011433$, $\nu = 1.296790$.

The drift correction w is imposed to ensure the martingale property under the risk–neutral measure $\mathbb{Q}$:

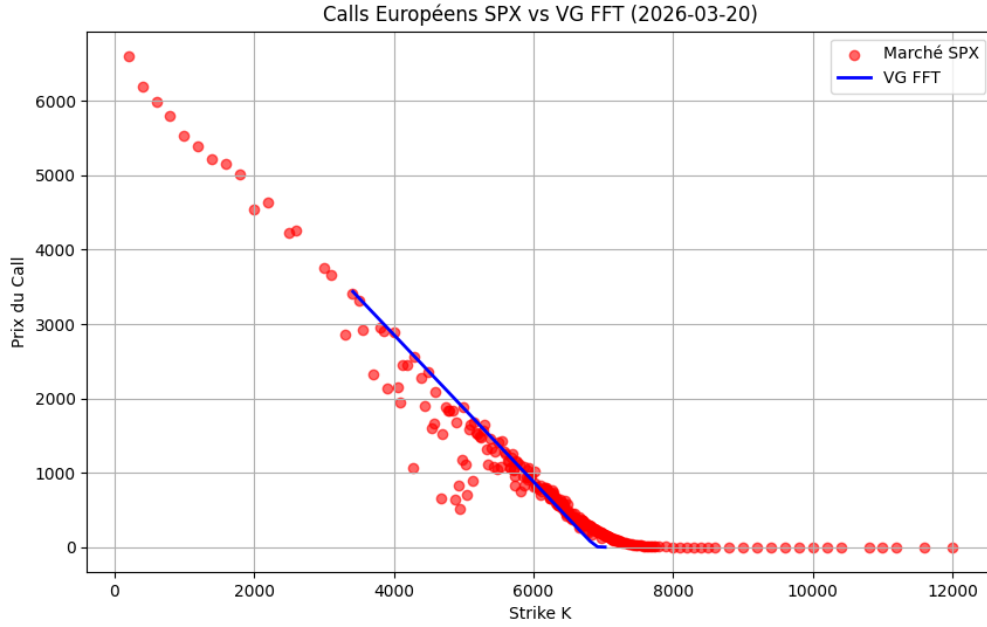
$$w = \ln \left[(1 - \theta\nu + \tfrac{1}{2}\sigma^2\nu)^{-1/\nu} \right].$$

To ensure convergence of the Fourier transform, we introduce a damping factor $\alpha = 1.2$. The strike grid is centred around S_0 , and the FFT parameters are:

$$N = 2^{14}, \quad \eta = 0.025.$$

The choice of η within a reasonable range (typically 0.01–0.05) ensures a sufficiently fine grid and numerical stability. Hence $\eta = 0.025$ is a methodological choice needed to properly capture the Fourier transform of the payoff.

6.2.3 Results



Interpretation The figure shows the prices of 6-month European call options computed using the Carr–Madan FFT method with the Variance–Gamma parameters calibrated via MLE. The strike interval relative to the spot price is: $0.5S_0$ to $1.5S_0$. Option prices decrease as the strike K increases, which is consistent with the expected behavior of European calls. The estimated option prices are consistent across the entire strike interval considered.

These results confirm that the MLE-calibrated Variance–Gamma model is suitable for pricing European options, effectively capturing tail behavior and return asymmetry observed in the market.

6.2.4 Summary: analysis of the European call and graphical interpretation

Reminder: A European call is a derivative instrument that grants its holder the right, but not the obligation, to buy an underlying asset at a predetermined strike price K at a fixed maturity T .

Intuitively, for a given strike K :

- The investor considers the probability that the future underlying price S_T is above K .
- If $S_T > K$, the call will be **in the money** at maturity, generating a positive payoff.
- In such a case, purchasing the call today may be beneficial.

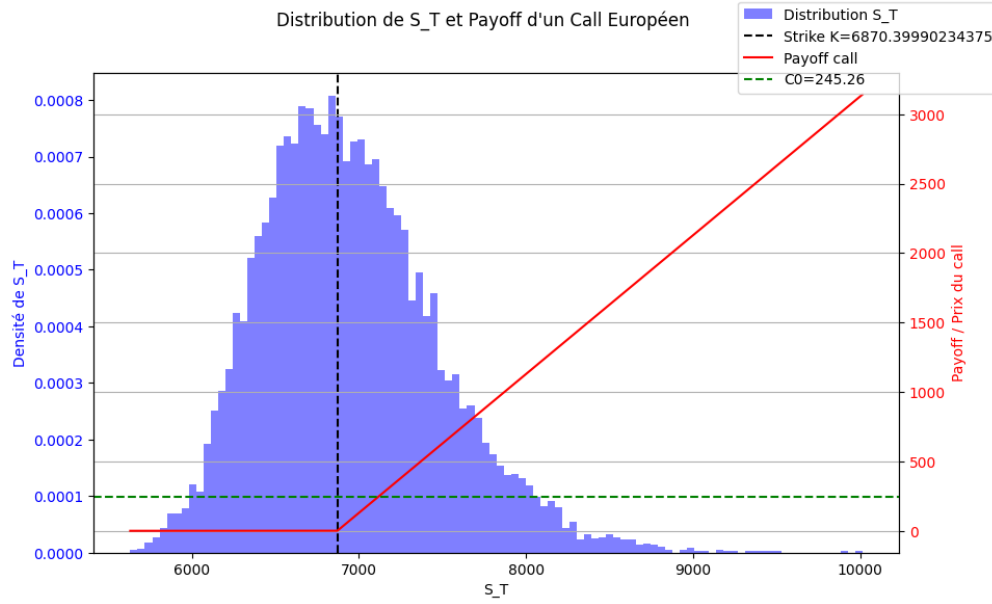
The Variance–Gamma (VG) model represents the probability distribution of S_T , incorporating heavy tails and asymmetry in market returns. The Carr–Madan FFT method provides an efficient tool to compute the current option price C_0 for all strikes, using the information embedded in the distribution of the underlying.

Comparing the model-implied price C_0 with the observed market price allows one to determine whether the option is:

- **Undervalued**, if the model price is lower than the market price;
- **Overvalued**, if the model price is higher than the market price;
- **Fairly valued**, if both prices are close.

Hence, the call represents a future right to buy the asset at K , and the present value C_0 corresponds to the fair amount to pay today for this right. If projections indicate that the option will likely end up in the money and C_0 remains reasonable, purchasing the call may be an attractive investment.

Graphical interpretation The accompanying figure combines the simulated distribution of S_T with the call payoff:



The model generated a European call price of 245.26 compared to a market price of 259.91, corresponding to approximately 5–6% below the observed market price. The simulated terminal distribution of S_T exhibited pronounced right skewness, with the strike located slightly above the central mass of the distribution. While the model captured asymmetric tail behavior and produced a realistic payoff structure, the pricing gap suggests that the calibrated parameters did not fully reproduce the market-implied right-tail risk. This divergence likely reflects either residual limitations in the jump specification (e.g., insufficient positive tail intensity) or the presence of risk premia embedded in option prices that are not captured by statistically estimated parameters.

7 Appendix A - Fundamental Derivations

7.1 A.1 Characteristic function of a compound Poisson process

- Consider a compound Poisson process:
 - The number of jumps $N \sim \text{Poisson}(\gamma)$.
 - The jump sizes X_j are i.i.d. with law $\mu(dx)$.

In a Lévy process, there is not a single type of event but a **Lévy intensity measure** ν distributed over different jump sizes x , corresponding to an infinite collection of localized (γ, μ) pairs.

- The characteristic function is:

$$\phi(u) = \mathbb{E}\left[e^{iu \sum_{j=1}^N X_j}\right] = \exp\left[\gamma \int_{\mathbb{R}} (e^{iux} - 1) \mu(dx)\right].$$

- Detail the calculation using conditional expectation on N :

$$\mathbb{E}\left[e^{iu \sum_{j=1}^N X_j}\right] = \sum_{n=0}^{\infty} \mathbb{E}\left[e^{iu \sum_{j=1}^N X_j} \mid N = n\right] \mathbb{P}(N = n).$$

- Conditional on $N = n$:

$$\mathbb{E}\left[e^{iu \sum_{j=1}^n X_j}\right] = \mathbb{E}\left[\prod_{j=1}^n e^{iu X_j}\right] = \prod_{j=1}^n \mathbb{E}[e^{iu X_j}] = (\mathbb{E}[e^{iu X_1}])^n.$$

- Recall that $\mathbb{P}(N = n) = e^{-\gamma} \frac{\gamma^n}{n!}$. Hence:

$$\mathbb{E}\left[e^{iu \sum_{j=1}^N X_j}\right] = \sum_{n=0}^{\infty} (\mathbb{E}[e^{iu X_1}])^n e^{-\gamma} \frac{\gamma^n}{n!} = e^{-\gamma} \sum_{n=0}^{\infty} \frac{(\gamma \mathbb{E}[e^{iu X_1}])^n}{n!}.$$

- Note: the exponential series satisfies

$$\sum_{n=0}^{\infty} \frac{\alpha^n}{n!} = e^{\alpha}, \quad \text{with } \alpha = \gamma \mathbb{E}[e^{iu X_1}].$$

- Finally we obtain:

$$\mathbb{E}\left[e^{iu \sum_{j=1}^N X_j}\right] = e^{-\gamma} e^{\gamma \mathbb{E}[e^{iu X_1}]} = \exp\left[\gamma (\mathbb{E}[e^{iu X_1}] - 1)\right] = \exp\left[\gamma \int_{\mathbb{R}} (e^{iux} - 1) \mu(dx)\right].$$

Remark: This formula shows that the characteristic function of a compound Poisson process is naturally expressed in terms of the Lévy measure $\nu(dx) = \gamma \mu(dx)$. It forms the basis for decomposing Lévy processes into jump components of different sizes.

7.2 A.2 Characteristic function of a normal distribution

- Let

$$X \sim \mathcal{N}(\mu, \sigma^2), \quad f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

- We want to compute the characteristic function:

$$\rho_X(u) = \mathbb{E}[e^{iux}] = \int_{\mathbb{R}} e^{iux} f_X(x) dx.$$

Thus,

$$\rho_X(u) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{+\infty} e^{iux} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) dx.$$

- Change of variable:

$$y = x - \mu, \quad x = \mu + y, \quad dx = dy.$$

Then:

$$\rho_X(u) = e^{iu\mu} \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{+\infty} \exp\left(iuy - \frac{y^2}{2\sigma^2}\right) dy.$$

- Complete the square:

$$iuy - \frac{y^2}{2\sigma^2} = -\frac{1}{2\sigma^2}(y^2 - 2iu\sigma^2 y) = -\frac{1}{2\sigma^2}[(y - iu\sigma^2)^2 - (iu\sigma^2)^2].$$

Therefore

$$e^{iuy - \frac{y^2}{2\sigma^2}} = \exp\left(-\frac{(y - iu\sigma^2)^2}{2\sigma^2}\right) \exp\left(\frac{(iu\sigma^2)^2}{2\sigma^2}\right) = \exp\left(-\frac{(y - iu\sigma^2)^2}{2\sigma^2}\right) \exp\left(-\frac{1}{2}u^2\sigma^2\right).$$

- The integral becomes:

$$\int_{-\infty}^{+\infty} e^{iuy} \exp\left(-\frac{y^2}{2\sigma^2}\right) dy = e^{-\frac{1}{2}u^2\sigma^2} \int_{-\infty}^{+\infty} \exp\left(-\frac{(y - iu\sigma^2)^2}{2\sigma^2}\right) dy.$$

The integral above is the integral of a Gaussian density centered at $iu\sigma^2$, whose value is simply $\sqrt{2\pi}\sigma$.

- Finally,

$$\rho_X(u) = \exp\left(iu\mu - \frac{1}{2}u^2\sigma^2\right).$$

Remark on the normalization factor. The factor $\sqrt{2\pi}\sigma$ appears because, even if the Gaussian center is translated in the complex plane ($y - iu\sigma^2$), the width and shape remain that of a normal density with variance σ^2 . The integral of a normal density over $\mathbb{R}$ is always $\sqrt{2\pi}\sigma$, regardless of its (real or complex) center. This property allows the direct derivation of the final result.

7.3 A.3 Characteristic function of the Gamma distribution

- The density of a Gamma variable with parameters (α, β) is:

$$f_G(x) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}, \quad x > 0, \alpha > 0, \beta > 0.$$

- In our case, we use the parametrization:

$$\alpha = \frac{1}{v}, \quad \beta = v.$$

- The characteristic function of G is:

$$\phi_G(u) = \mathbb{E}[e^{iuG}] = \int_0^{+\infty} e^{iux} f_G(x) dx.$$

Thus:

$$\phi_G(u) = \frac{1}{\Gamma(\alpha)\beta^\alpha} \int_0^{+\infty} e^{iux} x^{\alpha-1} e^{-x/\beta} dx.$$

- Combine exponentials:

$$e^{iux} e^{-x/\beta} = e^{-(\frac{1}{\beta} - iu)x}.$$

Let

$$A = \frac{1}{\beta} - iu, \quad \text{with } \Re(A) = \frac{1}{\beta} > 0.$$

The integral becomes:

$$\phi_G(u) = \frac{1}{\Gamma(\alpha)\beta^\alpha} \int_0^{+\infty} x^{\alpha-1} e^{-Ax} dx.$$

- Using the classical Euler integral property:

$$\Gamma(\alpha) = \int_0^{+\infty} x^{\alpha-1} e^{-x} dx, \quad \alpha > 0.$$

- Show that:

$$\int_0^{+\infty} x^{\alpha-1} e^{-Ax} dx = A^{-\alpha} \Gamma(\alpha).$$

- Substitution: $u = Ax$, $x = u/A$, $dx = du/A$:

$$\int_0^{+\infty} x^{\alpha-1} e^{-Ax} dx = \int_0^{+\infty} \left(\frac{u}{A}\right)^{\alpha-1} e^{-u} \frac{du}{A} = A^{-\alpha} \int_0^{+\infty} u^{\alpha-1} e^{-u} du = A^{-\alpha} \Gamma(\alpha).$$

- Deduce the characteristic function:

$$\phi_G(u) = \frac{1}{\Gamma(\alpha)\beta^\alpha} A^{-\alpha} \Gamma(\alpha) = (1 - iu\beta)^{-\alpha}, \quad \text{with } \Re(1 - iu\beta) > 0.$$

Remark: interpretation of the Gamma law in a Lévy process. In a Lévy process, the Gamma law often appears as a **waiting time distribution** in a Poisson process:

- If Poisson events occur at rate λ , the waiting time between two events is exponential with parameter λ .
- The time needed to observe α *independent events* then follows a $\text{Gamma}(\alpha, 1/\lambda)$ law.

In other words:

- α (shape) measures concentration: larger α yields a more concentrated distribution.
- β (scale) controls dispersion: it stretches or contracts the distribution.

7.4 A.4 Detailed derivation of the Carr Madan Fourier transform

We want to compute:

$$\psi_\alpha(v) = \int_{-\infty}^{\infty} e^{ivk} c_\alpha(k) dk.$$

Substitute

$$c_\alpha(k) = e^{\alpha k} C(e^k), \quad C(e^k) = e^{-rT} \mathbb{E}[(e^X - e^k)^+],$$

and using the density $f_X(x)$:

$$C(e^k) = e^{-rT} \left(\int_k^{\infty} e^x f_X(x) dx - e^k \int_k^{\infty} f_X(x) dx \right),$$

the transform becomes

$$\psi_\alpha(v) = e^{-rT} \int_{-\infty}^{\infty} e^{(iv+\alpha)k} \left[\int_k^{\infty} e^x f_X(x) dx - e^k \int_k^{\infty} f_X(x) dx \right] dk.$$

By interchanging the order of integration (Fubini's theorem, valid due to damping), we obtain two terms.

3.1 First term

$$\begin{aligned}
I_1 &= \int_{-\infty}^{\infty} e^{(iv+\alpha)k} \left(\int_k^{\infty} e^x f_X(x) dx \right) dk \\
&= \int_{-\infty}^{\infty} f_X(x) e^x \left(\int_{-\infty}^x e^{(iv+\alpha)k} dk \right) dx \\
&= \int_{-\infty}^{\infty} f_X(x) e^x \frac{e^{(iv+\alpha)x}}{iv+\alpha} dx \\
&= \frac{1}{iv+\alpha} \int_{-\infty}^{\infty} e^{(iv+\alpha+1)x} f_X(x) dx \\
&= \frac{\phi_T(v - i(\alpha+1))}{iv+\alpha},
\end{aligned}$$

where $\phi_T(u) = \mathbb{E}[e^{iuX}] = \int_{-\infty}^{\infty} e^{iuX} f_X(x) dx$ is the characteristic function of X , and $u = v - i(\alpha+1)$.

3.2 Second term

$$\begin{aligned}
I_2 &= \int_{-\infty}^{\infty} e^{(iv+\alpha)k} \left(-e^k \int_k^{\infty} f_X(x) dx \right) dk \\
&= - \int_{-\infty}^{\infty} f_X(x) \left(\int_{-\infty}^x e^{(iv+\alpha+1)k} dk \right) dx \\
&= - \int_{-\infty}^{\infty} f_X(x) \frac{e^{(iv+\alpha+1)x}}{iv+\alpha+1} dx \\
&= - \frac{\phi_T(v - i(\alpha+1))}{iv+\alpha+1}.
\end{aligned}$$

3.3 Combination of the two terms

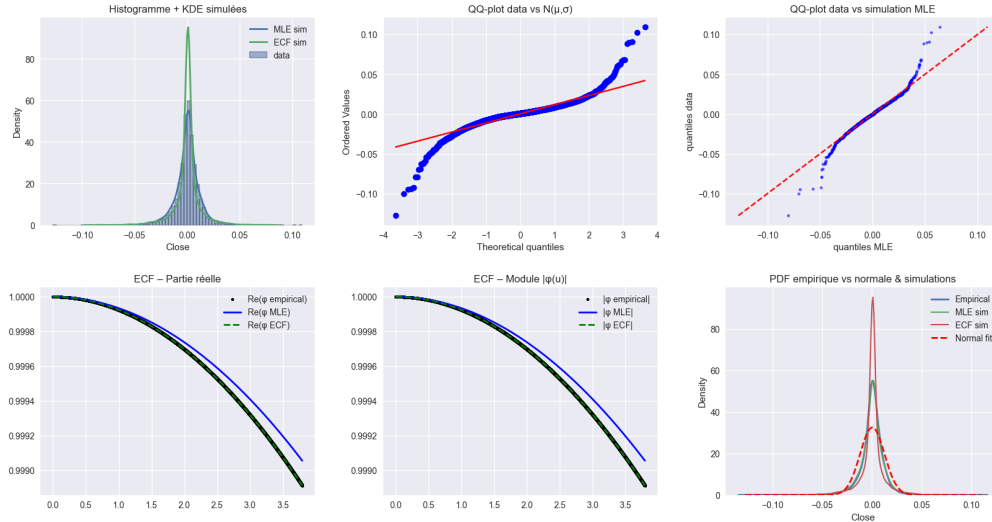
$$\begin{aligned}
\psi_{\alpha}(v) &= e^{-rT} \phi_T(v - i(\alpha+1)) \left(\frac{1}{iv+\alpha} - \frac{1}{iv+\alpha+1} \right) \\
&= e^{-rT} \phi_T(v - i(\alpha+1)) \frac{1}{(iv+\alpha)(iv+\alpha+1)}.
\end{aligned}$$

This corresponds to the standard Carr–Madan formula:

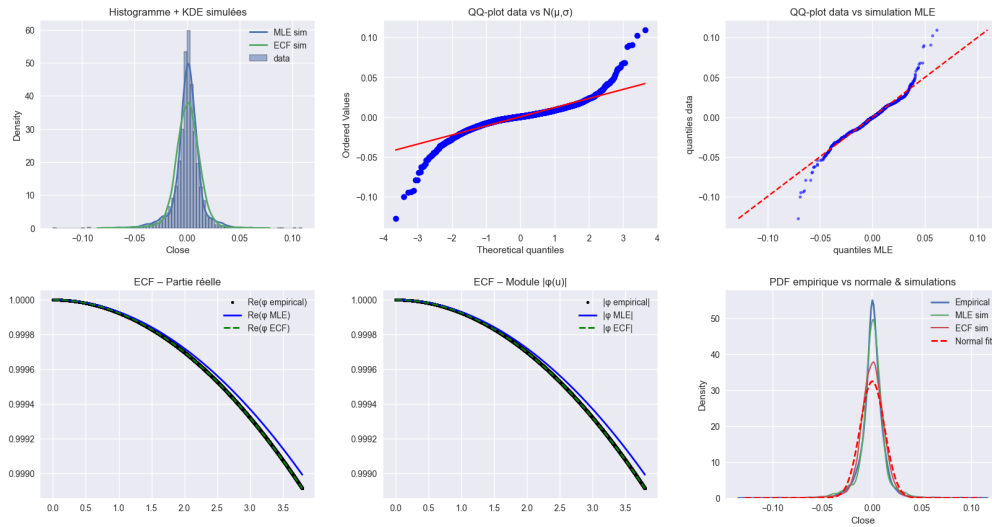
$$\psi_{\alpha}(v) = \frac{e^{-rT} \phi_T(v - i(\alpha+1))}{(iv+\alpha)(iv+\alpha+1)}.$$

8 Annexe B - Graphes

8.1 B.1 Comparaison of results for Variance Gamma



8.2 B.2 Comparaison of results for Merton



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